

Acquisition Parameter Detection Dataset

Task Description

This dataset was created for a binary classification task that differentiates PA chest X-ray images acquired using two different acquisition configurations. The negative class contains images acquired with a direct detector at 110 kVp, whereas the positive class contains images acquired with a scintillator detector at 120 kVp. The dataset is intended for the development and evaluation of a deep learning model that can distinguish between these acquisition configurations.

Labeling

Class labels were assigned using acquisition parameters extracted from the DICOM metadata headers. Images acquired with a direct detector at 110 kVp were assigned label 0, whereas images acquired with a scintillator detector at 120 kVp were assigned label 1.

Class Definitions

Class	Label	Description
Negative	0	PA chest X-ray images acquired with a direct detector at 110 kVp. These images were assigned label 0.
Positive	1	PA chest X-ray images acquired with a scintillator detector at 120 kVp. These images were assigned label 1.

Inclusion Criteria

Patients were required to satisfy the healthy clinical condition criteria. Only PA chest X-ray images with a known detector type, known tube voltage, and available exposure information were included. Eligible images were acquired either with a direct detector at 110 kVp or with a scintillator detector at 120 kVp. Exposure was required to be greater than 0 mAs and less than 6.9 mAs.

Exclusion Criteria

Images were excluded if the patient did not satisfy the healthy clinical condition criteria, the image view was not PA, or the required acquisition parameters were missing or unknown. Images were also excluded if the detector type or tube voltage did not match one of the two target acquisition configurations, or if the exposure value was outside the specified range. Unreadable or unsuccessfully converted DICOM images were excluded during dataset generation.

Dataset Split Statistics

Split	Negative Patients	Positive Patients	Negative Images	Positive Images
Training	993	993	2803	993
Validation	2019	331	2019	418

Split	Negative Patients	Positive Patients	Negative Images	Positive Images
Test	2019	331	2494	877
Total	5031	1655	7316	2288

Processing Errors

Failed images are excluded from the successful image counts and from the generated CSV files.

Split	Class	Failed Images
Training	Negative	0
Training	Positive	0
Validation	Negative	0
Validation	Positive	0
Test	Negative	0
Test	Positive	0
Total	All classes	0

No processing errors were recorded.

Dataset Summary

Metric	Negative	Positive	Total
Patients	5031	1655	6686
Successfully saved images	7316	2288	9604
Processing errors	0	0	0
Attempted images	7316	2288	9604

Overall processing error rate: 0.00%.